

Ethanol Blending in Gasoline – Spain

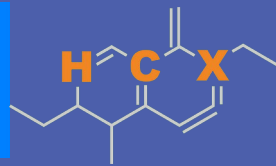
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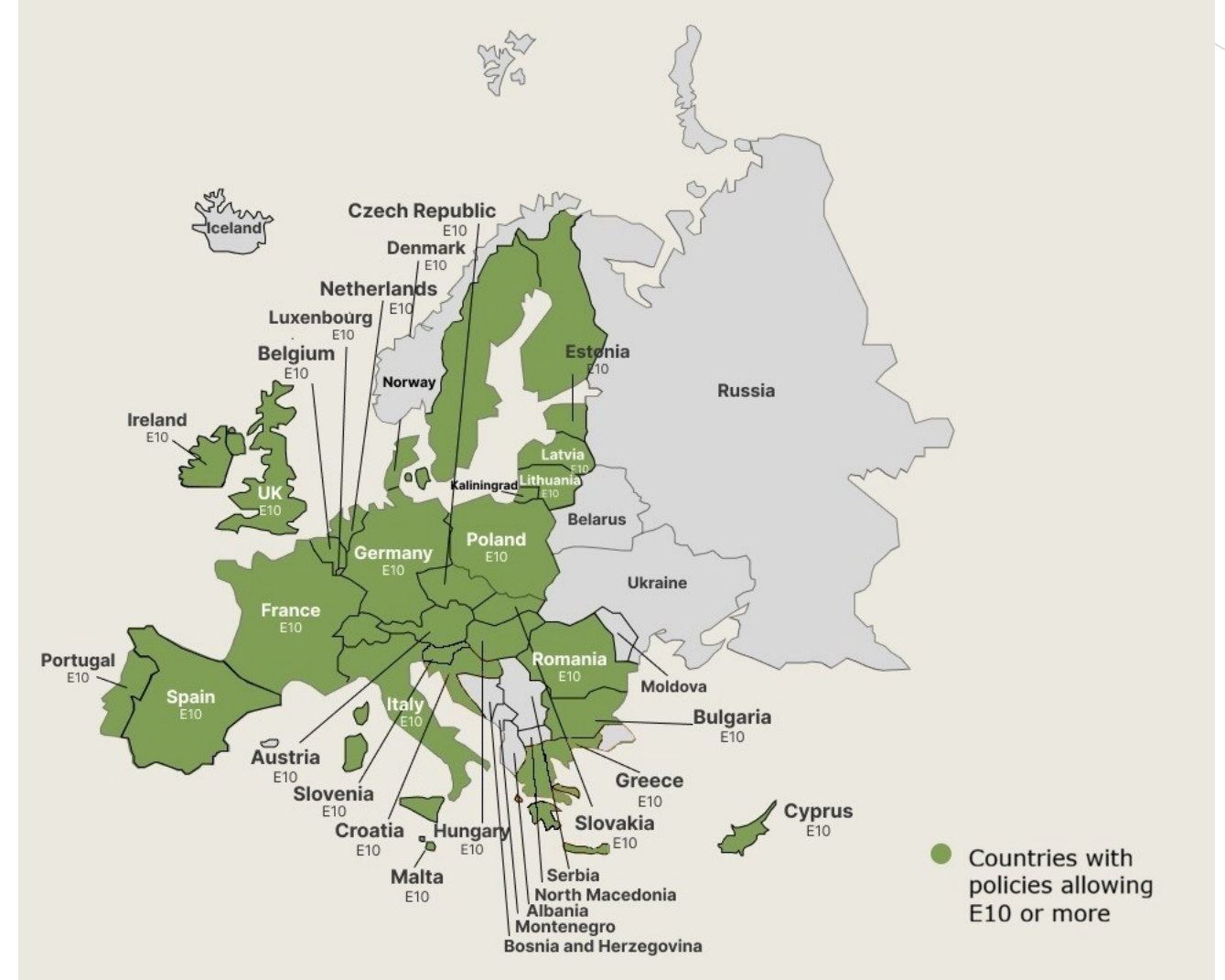


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

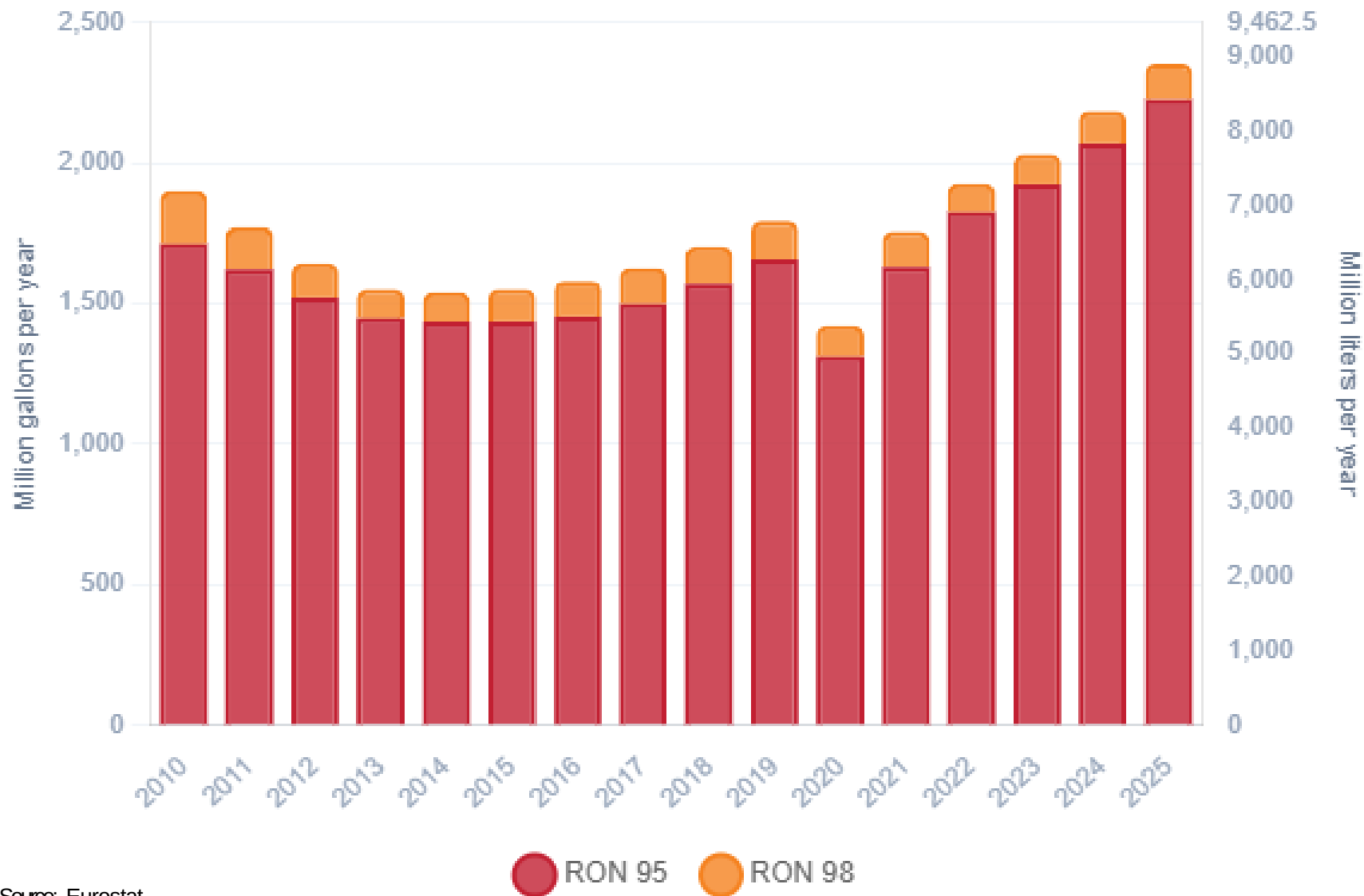
Ethanol Gasoline Blending in Spain



In 2025, gasoline consumption in Spain was 8,903 million liters (2,352 million gallons) and growth rate was 1.4%. Gasoline production and net imports were 10,808 and -1,906 million liters (2,855 and -503 million gallons) correspondingly. Spain complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

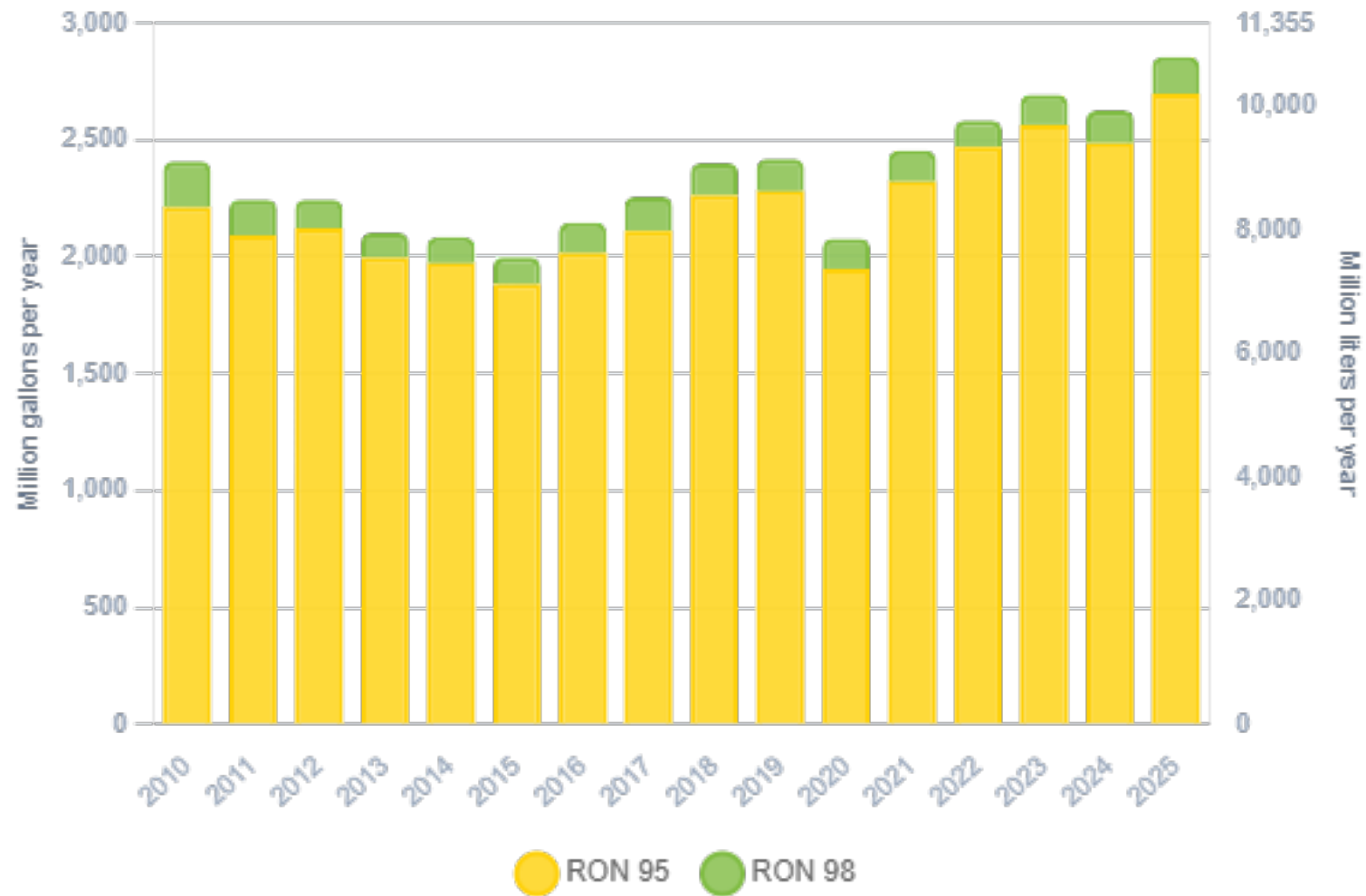
In 2025, ethanol consumption in Spain was 315 million liters (83 million gallons) representing 3.5% of gasoline demand. Ethanol production and Net Imports were 540 and -333 million liters (143 and -88 million gallons) correspondingly. Ethanol sources are Corn 60%, Cereals 25% and Sugar Beet/Wine 40%.

Gasoline Demand in Spain



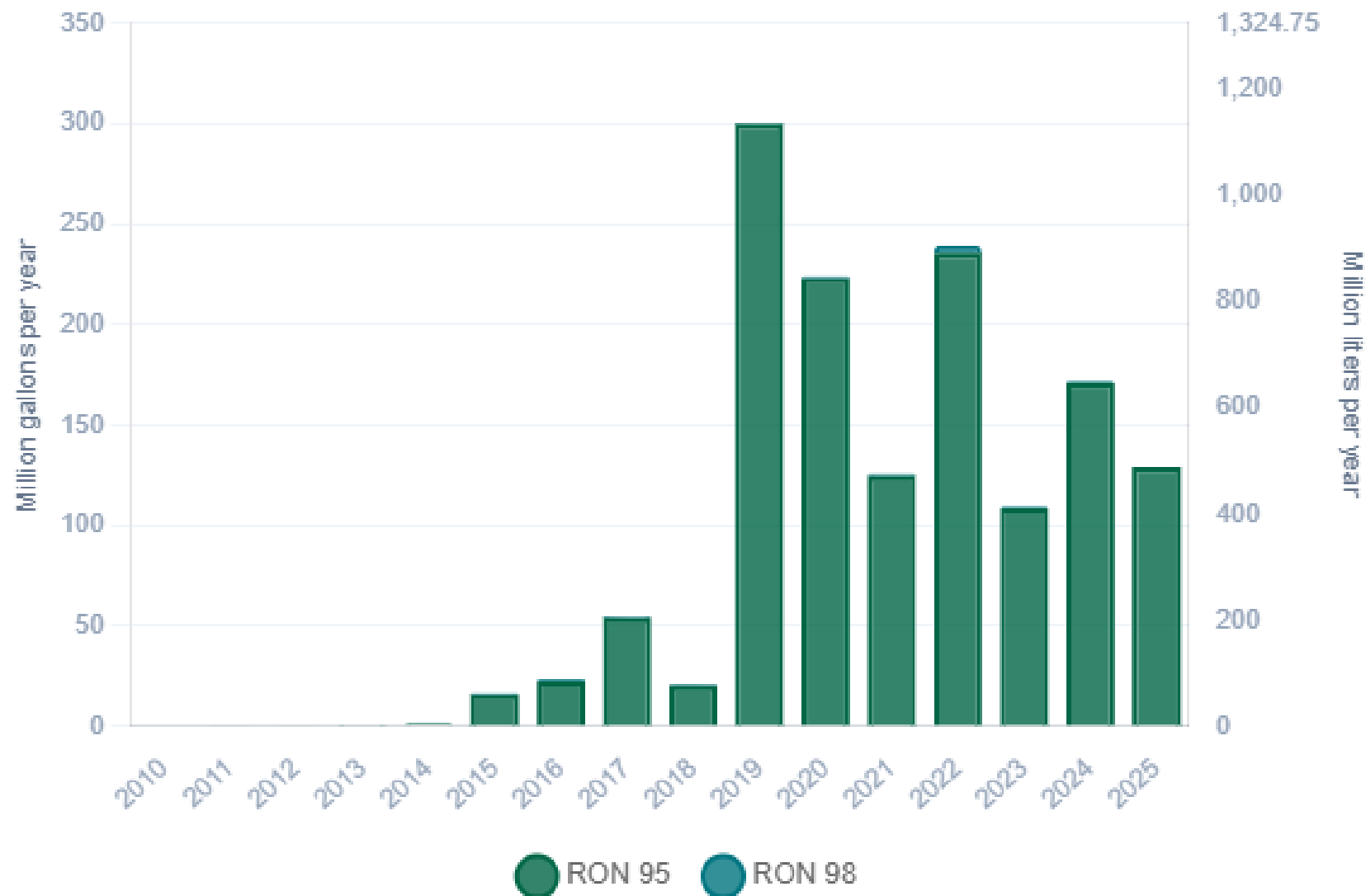
Source: Eurostat

Gasoline Production in Spain



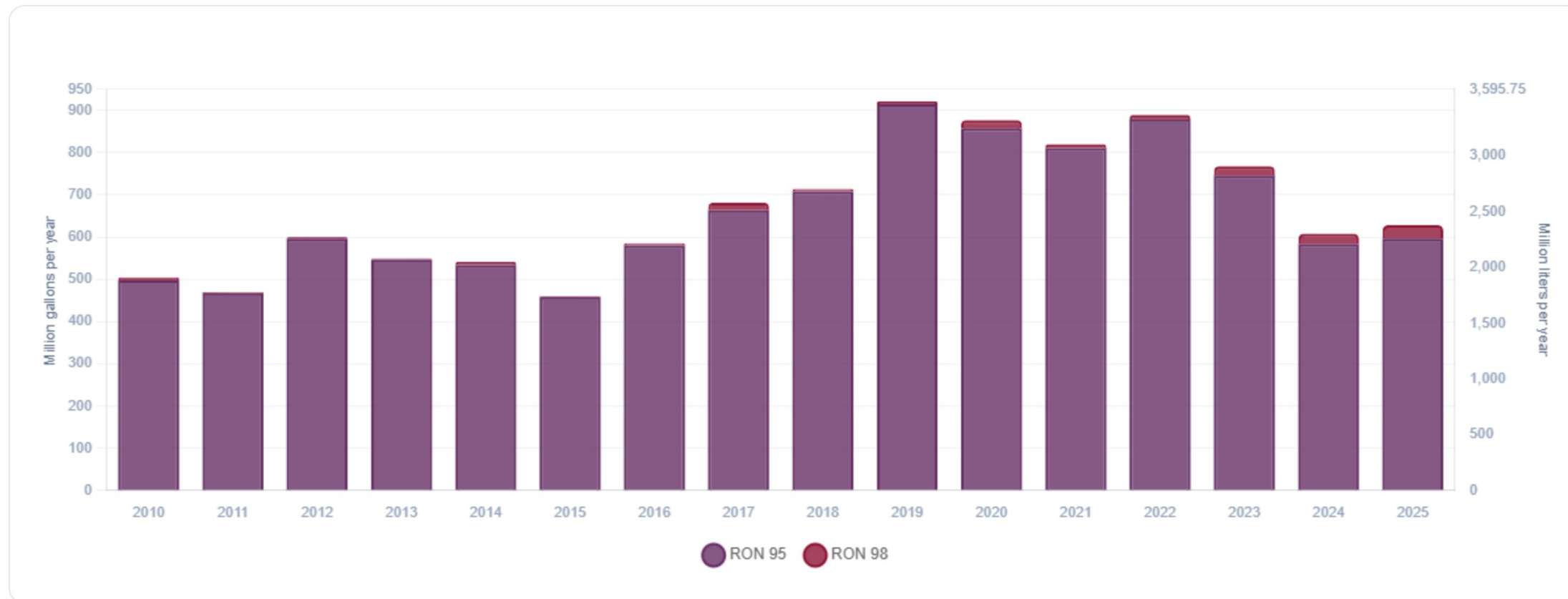
Source: Eurostat

Gasoline Imports in Spain



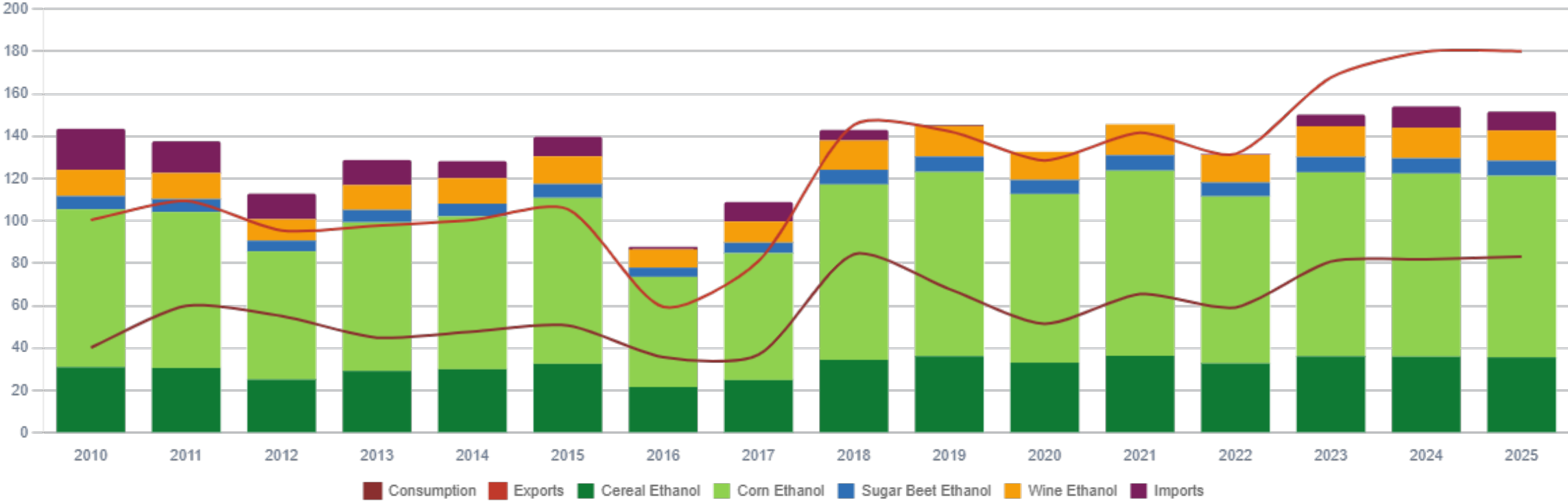
Source: Eurostat

Gasoline Exports in Spain



Source: Eurostat

Ethanol Balance in Spain



Source: Eurostat

Gasoline Quality in Spain

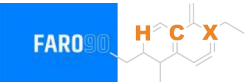


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.5							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

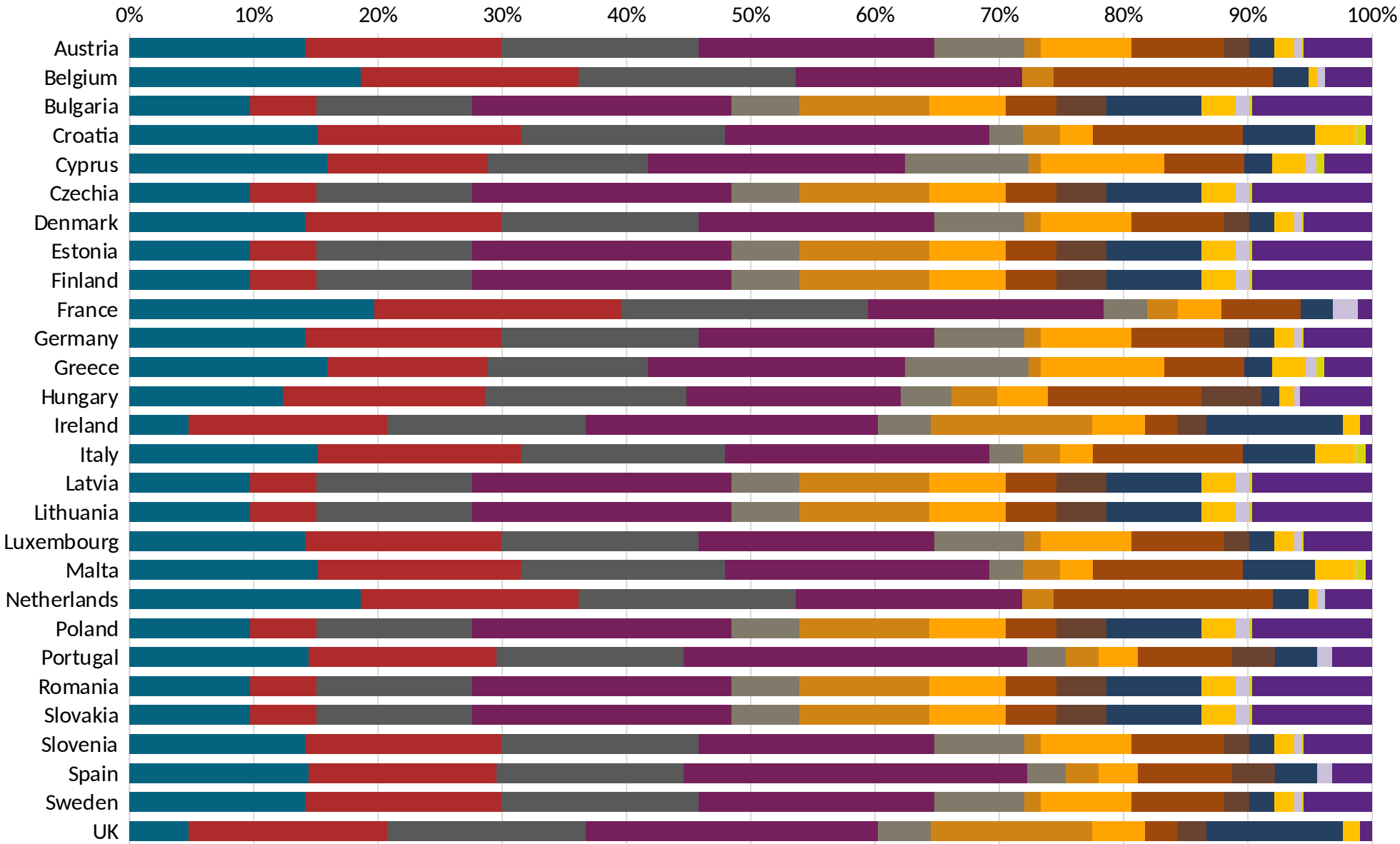
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



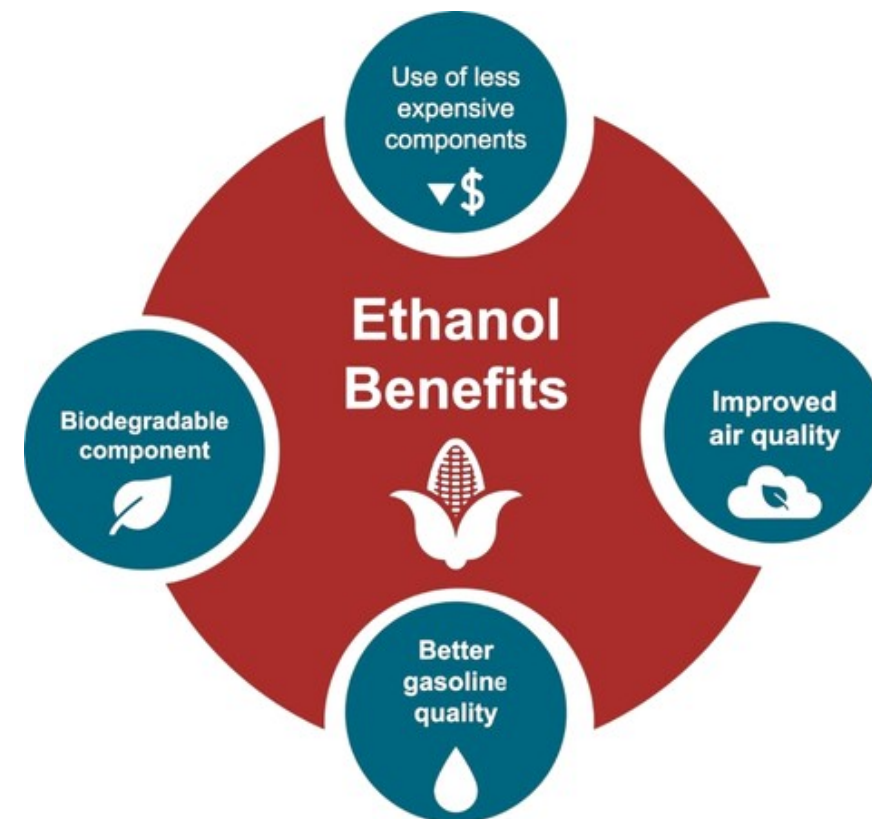
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

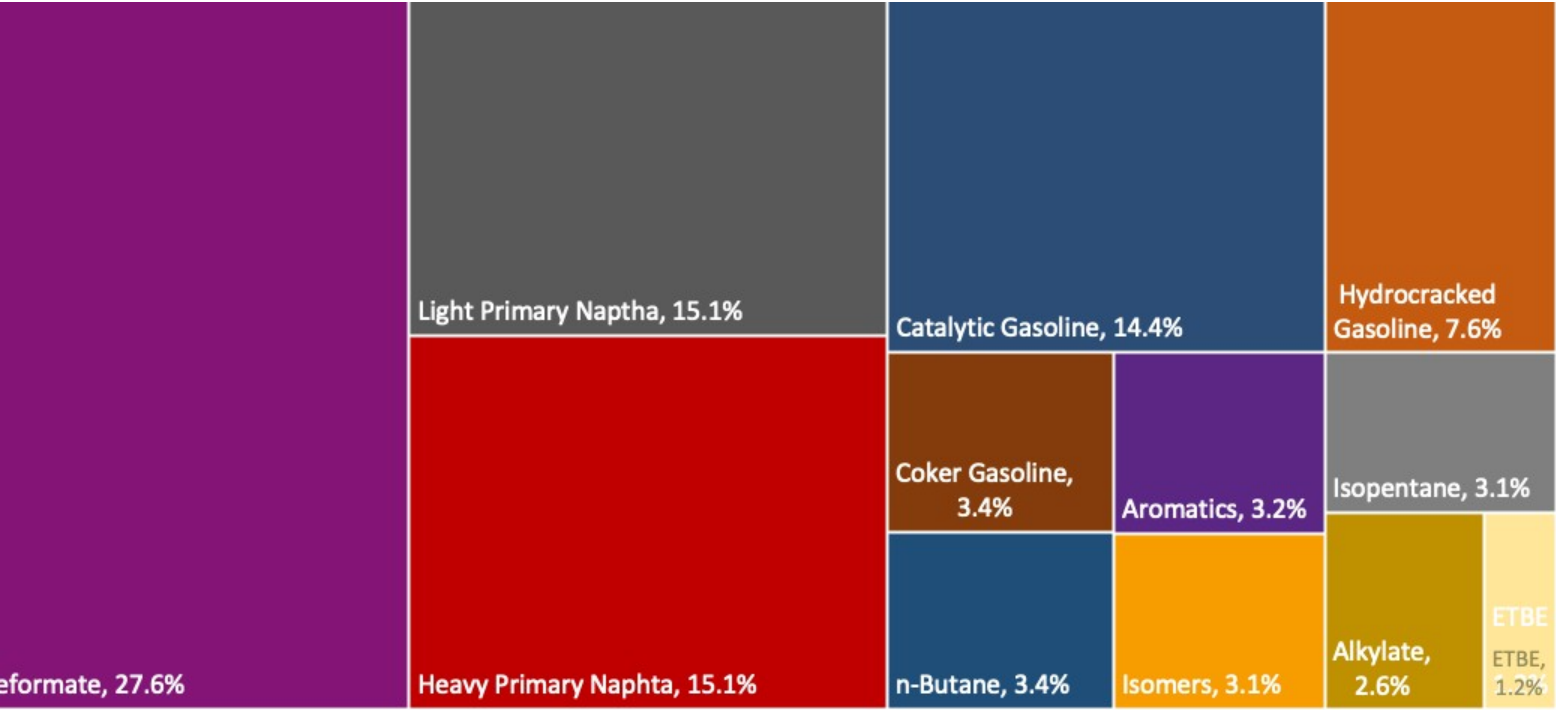
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

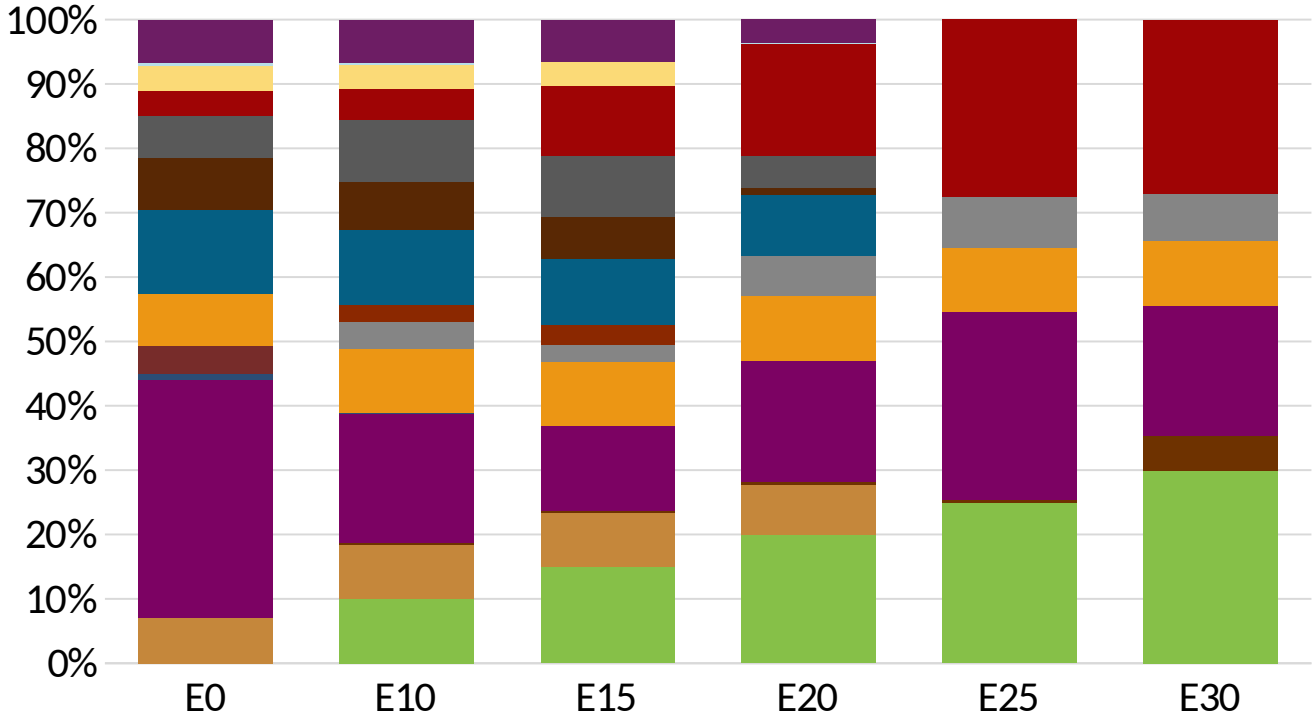
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Spain Available Blending Components

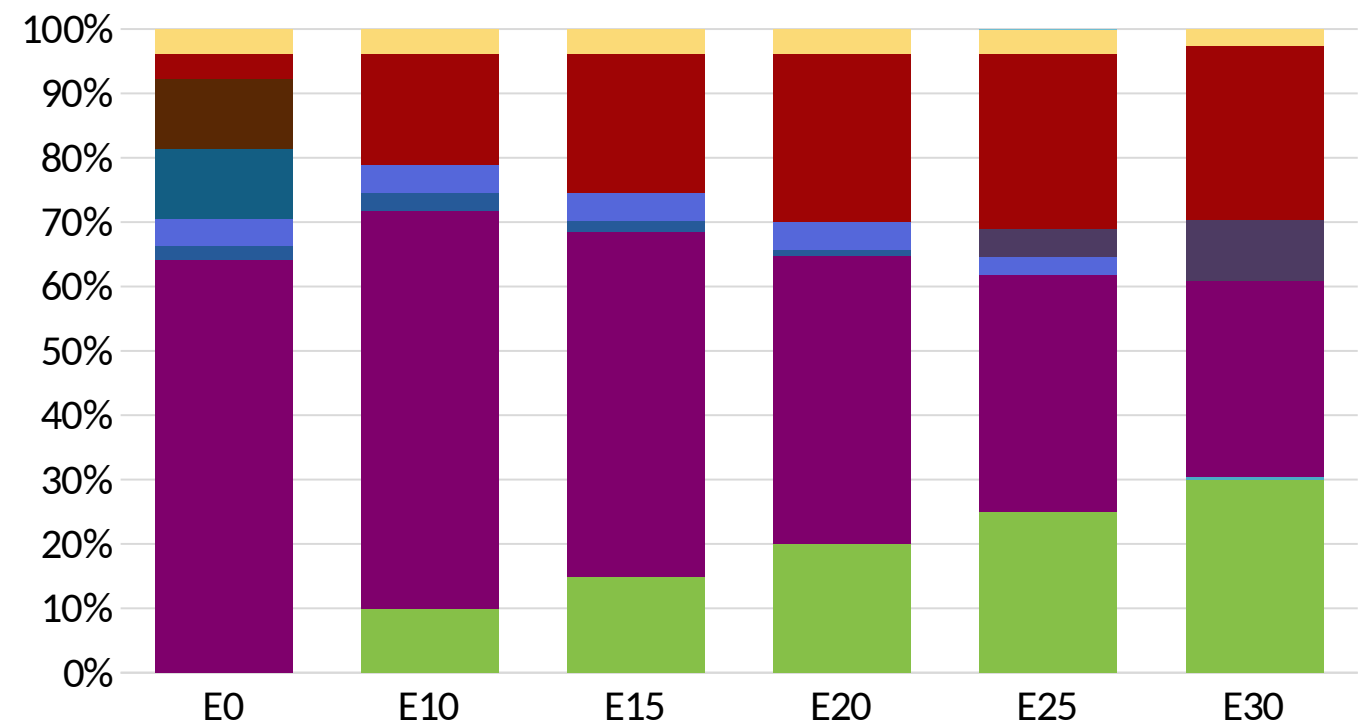


Spain – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

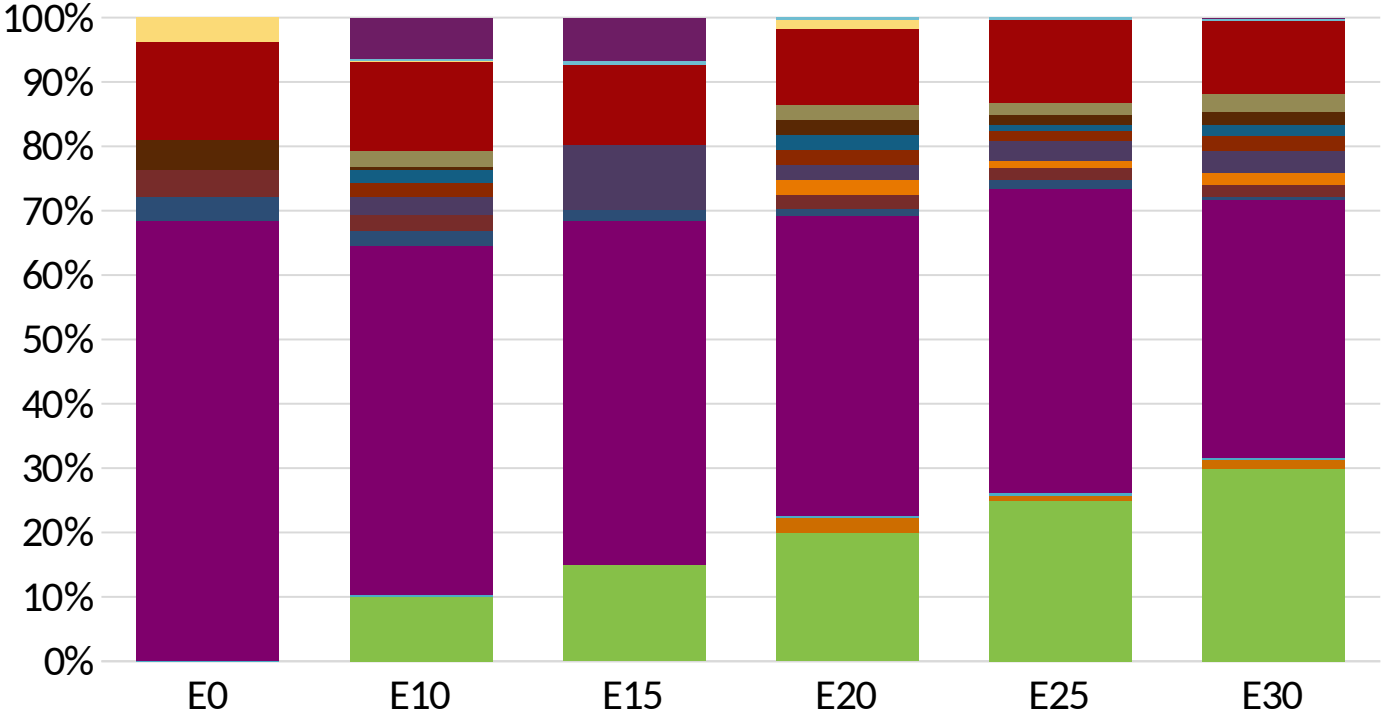
Spain - Gasoline 98 - Constant Octane



Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.146	\$ 2.022	\$ 1.953	\$ 1.878	\$ 1.800	\$ 1.721

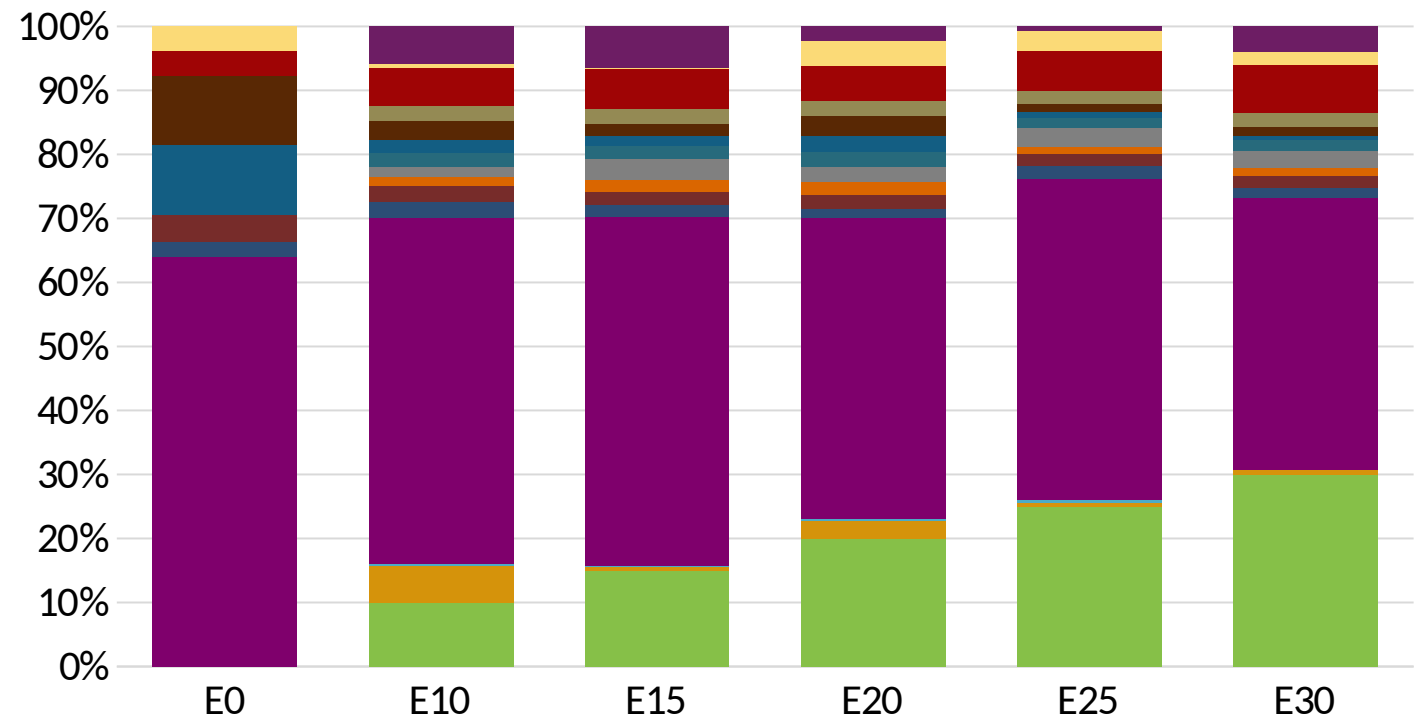
Spain – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	95.0	96.8	98.2	99.9	101.4	102.6
Price (US\$/gal)	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009

Spain – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	99.3	100.9	102.5	104.0	105.0
Price (US\$/gal)	\$ 2.146	\$ 2.146	\$ 2.146	\$ 2.146	\$ 2.146	\$ 2.146

Source:
Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

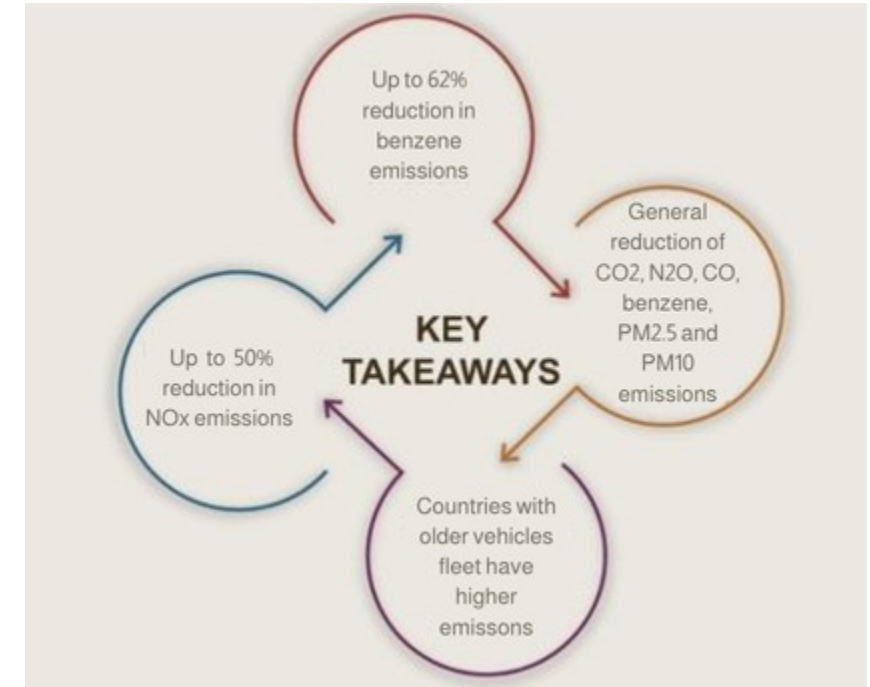
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

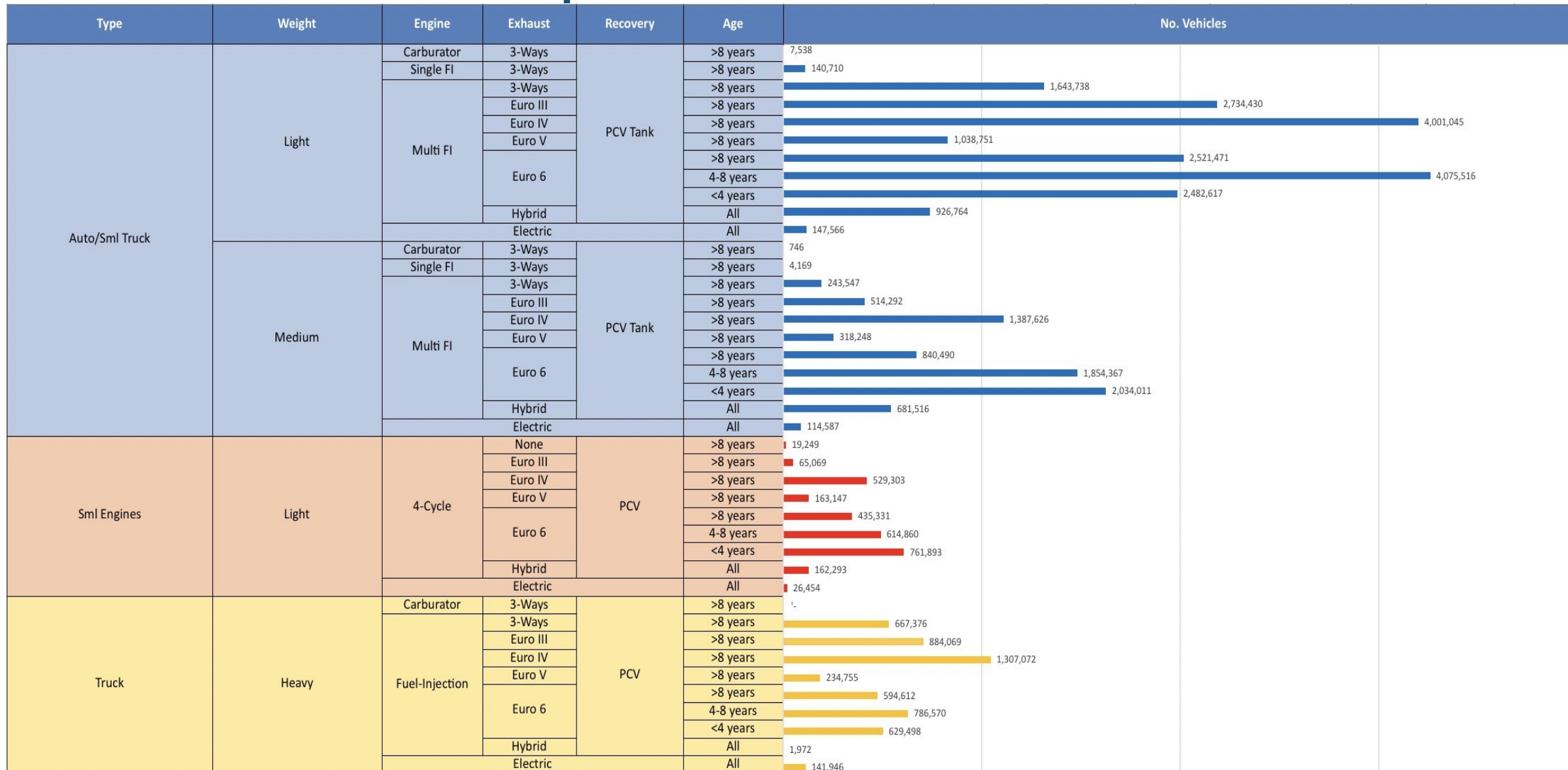
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Spain

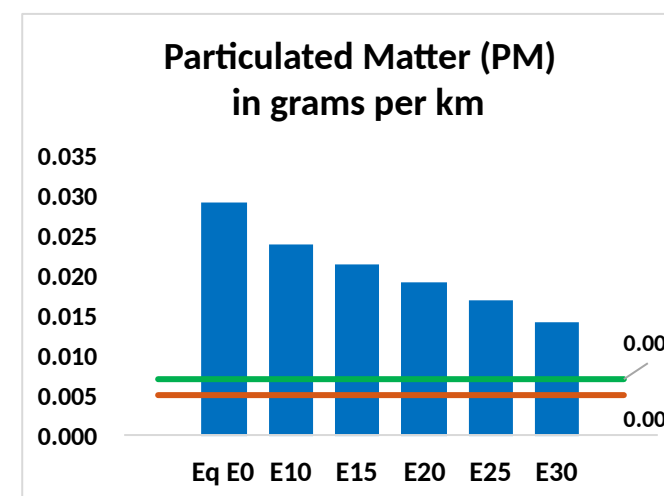
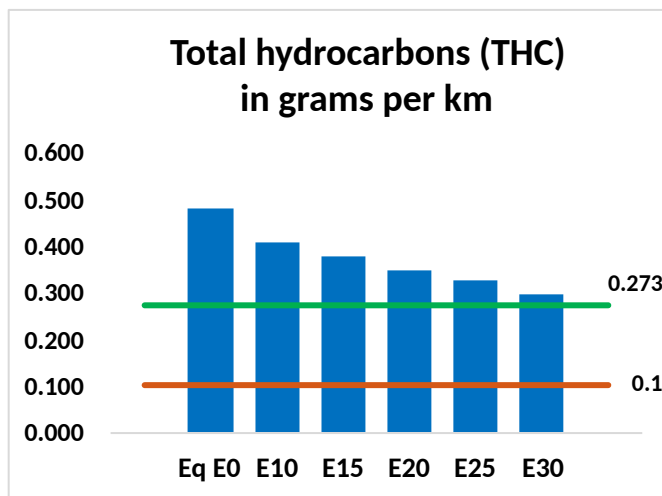
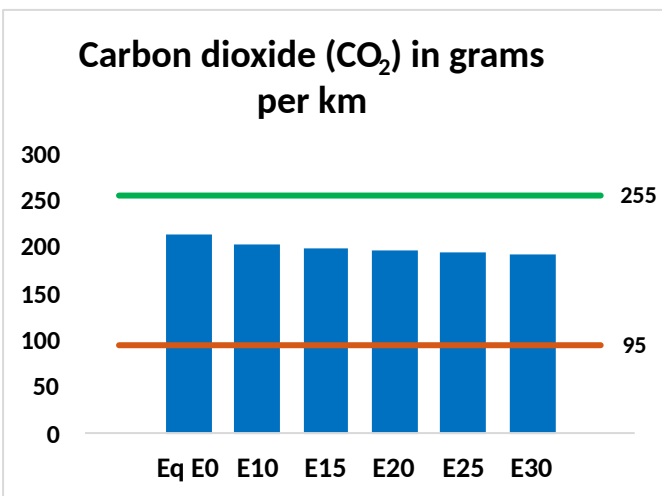
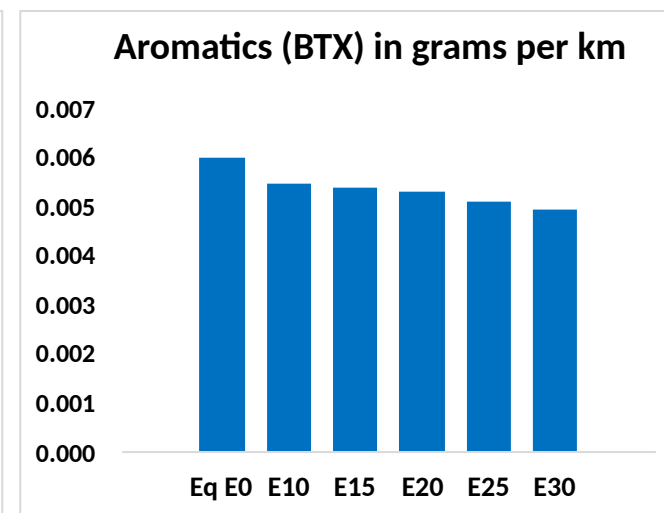
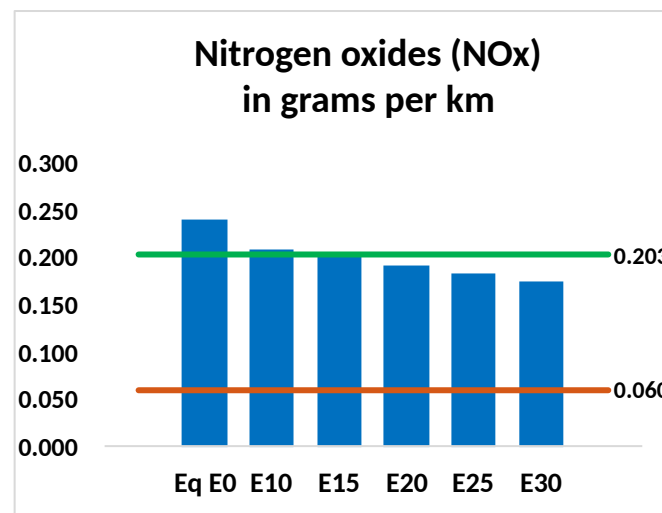
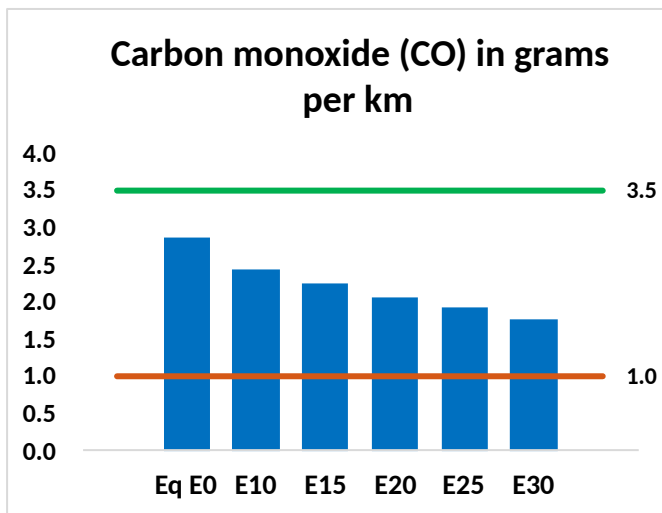
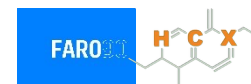


Vehicle Fleet: 35,739,214
Gasoline Vehicle Fleet: 11,668,578

Source: Eurostat, ACEA

Spain – Vehicular Emissions

— TIER III BIN 125 United States
— Euro 6 Standard



Spain – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	844,467	778,638	712,809	658,688	604,142	562,175	516,809	-16%	-28%
CO2	62,637,573	61,026,893	59,505,695	58,309,317	57,717,302	57,380,467	56,322,780	-5%	-8%
VOC	140,959	130,441	119,923	111,366	102,784	96,193	87,367	-15%	-27%
NOx	70,807	65,378	61,657	59,762	56,709	54,316	51,498	-13%	-20%
SOx	712	659	605	559	515	468	418	-15%	-28%
NH3	22,136	21,917	21,698	21,801	22,046	22,219	22,362	-2%	0%
N2O	1,043	1,037	1,033	1,038	1,059	1,071	1,083	-1%	2%
CH4	29,799	29,799	29,799	30,245	30,901	31,467	32,004	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	816	786	755	737	721	710	692	-7%	-12%
Acetaldehyde	2,342	2,688	3,944	6,006	8,183	9,351	11,049	68%	249%
Formaldehyde	8,410	8,177	9,532	11,192	11,726	12,972	14,121	13%	39%
Benzene	1,765	1,696	1,607	1,582	1,568	1,500	1,452	-9%	-11%
PM2.5	3,664	3,331	2,998	2,705	2,418	2,111	1,789	-18%	-34%
PM10	4,939	4,490	4,041	3,646	3,260	2,845	2,411	-18%	-34%

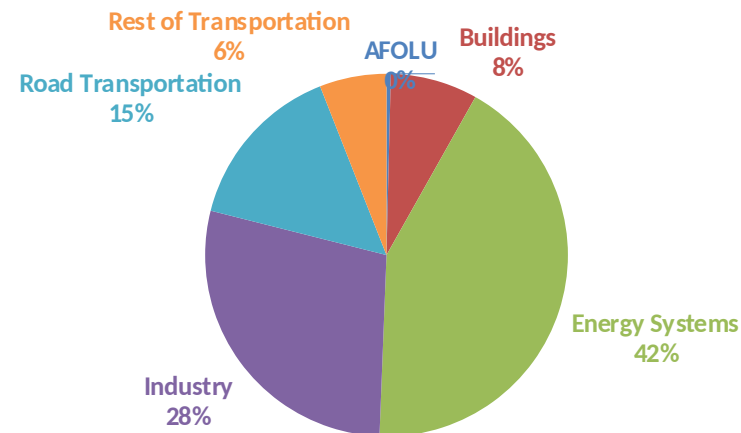
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

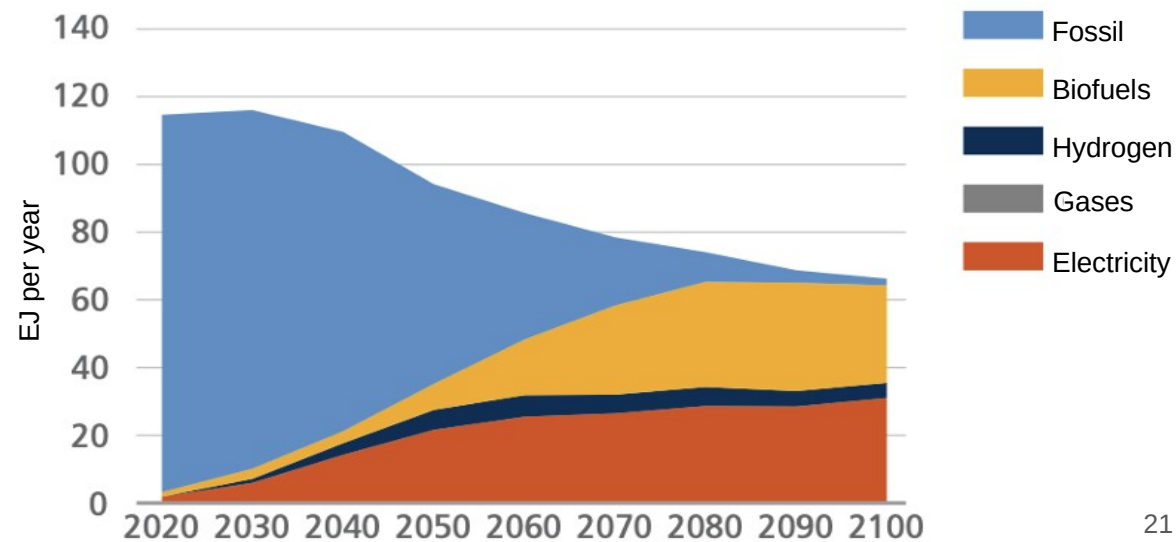
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

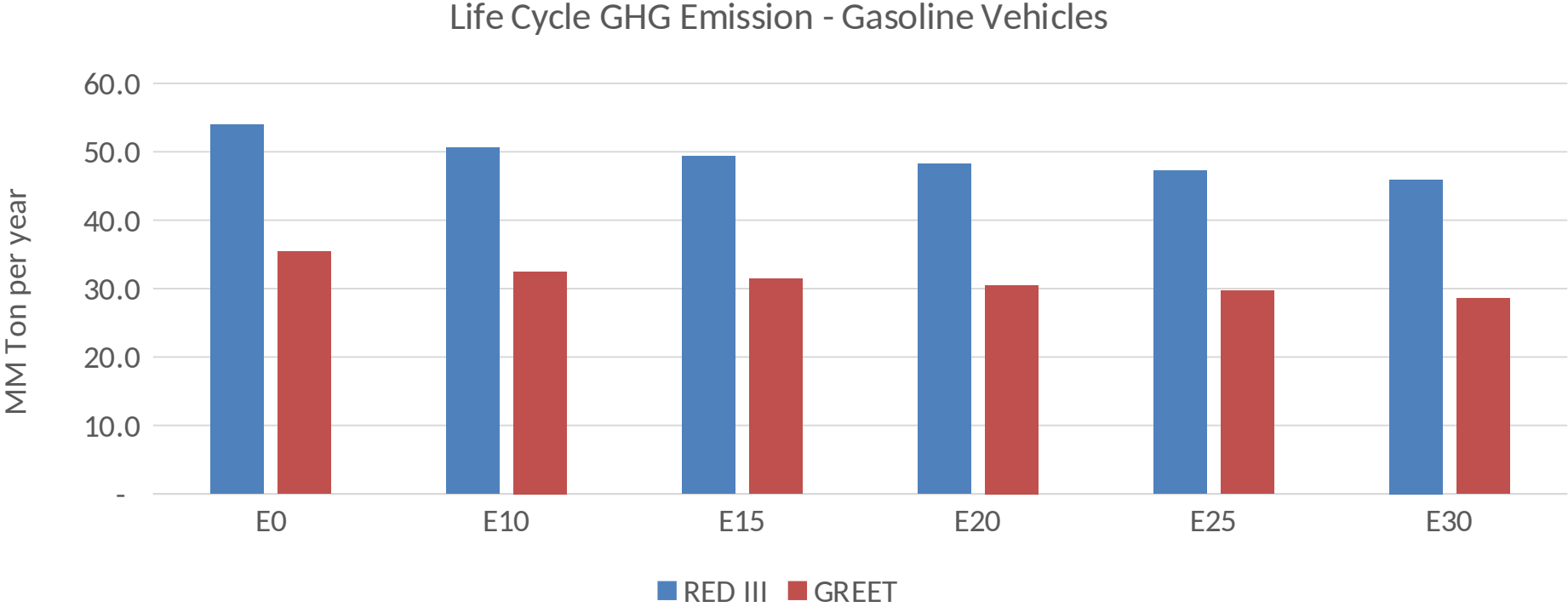
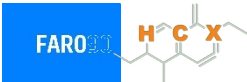
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Spain – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	269.9
Target @ 2035 (MMT/year)	151.9
Delta	118.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.2%	11.2%	13.9%	16.1%	19.3%
RED II/III	0.0%	6.3%	8.6%	10.7%	12.5%	14.9%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.5%	3.3%	4.2%	4.8%	5.8%
RED II/III	0.0%	2.9%	3.9%	4.9%	5.7%	6.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90